

Understanding Urban Heat Islands in Dhaka City Through Explainable GeoAI

Maria Sultana

Florida State University

Md Didarul Islam

`mddidaruli@uidaho.edu`

University of Idaho

Bradford Johnson

Florida State University

Research Article

Keywords:

Posted Date: September 15th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-7547419/v1>

License:   This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Understanding Urban Heat Islands in Dhaka City Through Explainable GeoAI

Maria Sultana^a, Md Didarul Islam^{b*}, Bradford Johnson^a

^a Department of Geography, Florida State University

^b Department of Forest, Rangeland and Fire Sciences, University of Idaho

* Corresponding author email: mddidaruli@uidaho.edu

Understanding Urban Heat Islands in Dhaka City Through Explainable GeoAI

Abstract

Urban Heat Islands (UHIs) are intensifying across rapidly urbanizing megacities in the Global South, posing serious threats to public health, infrastructure, and environmental equity. Understanding the drivers of urban heat is particularly challenging due to the nonlinear and spatially heterogeneous nature of land surface temperature (LST) dynamics. This study presents the first application of a spatially explainable GeoAI framework to analyze LST variation across Dhaka, one of the most densely populated and climate-vulnerable megacities in the Global South. Leveraging multi-source geospatial and remote sensing data, a 500 m grid-based machine learning model was developed using AutoML (FLAML) with LightGBM as the selected estimator. To enhance interpretability, the study employed both SHAP and GeoShapley methods to quantify global feature importance and spatially varying effects, enabling a critical comparison of their explanatory capacities in capturing geographic heterogeneity. Results reveal that built-up intensity, bare surfaces, and vegetation structure significantly influence urban thermal patterns, with strong spatial heterogeneity in their effects. GeoShapley decomposition highlights localized cooling thresholds for green and blue infrastructure and identifies spatial clusters of intrinsic thermal drivers not explained by observed variables. This integrated approach not only improves predictive performance but also supports spatially targeted heat mitigation strategies. The proposed framework is scalable and transferable, offering a practical template for climate-resilient urban planning in other data-constrained cities.

Introduction

Rapid urbanization is reshaping the Earth's surface by converting vegetated and pervious land into impervious materials such as concrete and asphalt. These surfaces have a high thermal capacity, causing them to absorb and retain heat, thereby intensifying the UHI effect and increasing LST in urban areas (Bappa et al., 2022; Kim et al., 2022; Kulsum & Moniruzzaman, 2022; Voogt & Oke, 2003). The consequences of the UHI effect are multifaceted, including heightened public health risks, increased energy consumption for cooling, and the deterioration of urban ecological systems (Cleland et al., 2023; Feng et al., 2024; Foroutan et al., 2025; Wei et al., 2025). These impacts are particularly severe in rapidly growing tropical megacities, where existing socio-economic

vulnerabilities exacerbate the adverse effects of rising urban temperatures (Abrar et al., 2022; Tabassum et al., 2025).

The UHI results from complex, nonlinear, and spatially heterogeneous processes influenced by a range of factors, including land use, vegetation cover, building morphology, and atmospheric conditions (Bappa et al., 2022; Imran et al., 2021; Morshed et al., 2017; Uddin et al., 2022). Numerous studies have sought to understand, model, and mitigate UHI effects using both statistical and machine learning approaches (R. Chen et al., 2024; Imran et al., 2021; Islam et al., 2021; Islam et al., 2019; Tabassum et al., 2025). Traditional methods such as linear regression, and geographically weighted regression have been instrumental in identifying general spatial trends (Si et al., 2024). However, these techniques often fall short in capturing nonlinear interactions and spatially varying thresholds (Si et al., 2024). Moreover, their predictive capabilities are limited, particularly in data-scarce environments like Dhaka, where localized and actionable insights are critical for effective climate resilience planning (Islam, et al., 2022a; Islam, et al., 2022b; Li, 2024).

In contrast, while statistical models can incorporate diverse datasets, machine learning algorithms offer notable advantages in handling large-scale, high-dimensional data and capturing complex nonlinear relationships. These capabilities allow ML models to achieve superior predictive accuracy and integrate heterogeneous sources such as satellite-derived indices, urban morphology, and land use patterns (Hu et al., 2023; Islam et al., 2021; Kim et al., 2022; Pande et al., 2024). Yet, despite their advantages, these models are often criticized as "black boxes" due to their lack of explainability, which hinders interpretation and reduces trust among urban planners and policymakers (Islam, et al., 2022a; Islam, et al., 2022b; Rihan et al., 2025; Sheng et al., 2025). This has led to the growing adoption of Explainable Artificial Intelligence (XAI) techniques, particularly the SHapley Additive exPlanations (SHAP) framework (Lundberg & Lee, 2017a). Rooted in cooperative game theory, SHAP attributes a model's prediction to individual input features, providing a transparent way to quantify each variable's contribution. SHAP has proven useful across a wide range of domains for interpreting complex models and supporting data-driven inference (Feng et al., 2024; Hu et al., 2023). However, a key limitation is that SHAP assumes observations are independent and identically distributed (i.i.d.), ignoring spatial dependence as established in Tobler's First Law - the tendency for nearby areas to exhibit similar values due to shared environmental or contextual factors (Tobler, 1970). As Li (2024) emphasizes, this

limitation makes SHAP less suitable for geospatial applications, as it cannot distinguish between the influence of a location's physical or environmental features and the effects arising solely from its geographic position (Li, 2024). As a result, SHAP may misattribute variations in the predicted outcome to observed features, without recognizing the influence of underlying spatial structure that also contributes to those patterns (Li, 2024).

To overcome these limitations, recent advancements in geospatial artificial intelligence (GeoAI) have introduced spatially-aware explainability tools. Among them, GeoShapley - a novel spatial extension of SHAP developed by Li (2024) has emerged as a powerful model-agnostic framework (Li, 2024). GeoShapley decomposes predictions into three interpretable components: intrinsic location effects, feature effects, and spatial-feature interactions. This allows for spatially explicit interpretation of how variables influence outcomes, aligning conceptually with geographically weighted regression but offering greater flexibility and scalability. While GeoShapley has been applied in fields such as public health (Foroutan et al., 2025), transportation (Peng et al., 2025), carbon emissions (R. Chen et al., 2024; Wu et al., 2025), housing prices (Y. Chen et al., 2025), and flood risk (E. Ke et al., 2025), its use in modeling urban heat remains limited particularly in Global South megacities like Dhaka, where spatial equity and climate vulnerability are pressing concerns.

Dhaka, Bangladesh, one of the most densely populated megacities globally, serves as a striking example of the urban heat challenges intensified by rapid, unplanned urbanization (Kafy et al., 2021; Morshed et al., 2017). The city has undergone dramatic land cover transformations over the past few decades, driven by soaring population growth and unmanaged development. Between 1990 and 2020, the impervious surface area in Dhaka more than tripled, increasing by approximately 215%, while vegetative cover experienced a substantial decline (Uddin et al., 2022). Satellite-based analyses have documented an average rise in LST of $2.4^{\circ}\text{C} \pm 0.8^{\circ}\text{C}$, with peak temperatures exceeding 38°C in informal settlements during the pre-monsoon season (Imran et al., 2021; Rahman et al., 2020; Tabassum et al., 2025; Uddin et al., 2022). Despite these observed trends, systematic assessments of vegetation loss and its long-term thermal consequences remain limited in the literature. Furthermore, mesoscale climate modeling reveals that Dhaka's dense built environment and disrupted wind circulation amplify the spatial heterogeneity of heat exposure across the city (Tabassum et al., 2025). These thermal stresses disproportionately affect low-

income communities, particularly those living in informal settlements with limited access to adaptive infrastructure such as green spaces, natural ventilation, or cooling technologies (Abrar et al., 2022; Ahmed et al., 2013).

This study addresses a critical research gap by integrating explainable GeoAI to unravel the spatial complexity of urban heat dynamics in Dhaka. First, it presents the application of GeoShapley to model and interpret UHI patterns in the city, offering a spatially explicit understanding of how urban features such as vegetation, built-up intensity, and morphology interact with local land surface temperatures. Unlike traditional models, this approach captures the geographically varying influence of each variable, allowing for nuanced interpretation beyond global averages. Second, the study identifies location-specific thresholds and interactions, such as the minimum green space required to achieve measurable cooling in densely populated neighborhoods providing a data-driven basis for equitable and context-sensitive heat mitigation. Third, it develops a novel GeoAI-based analytical framework - the first of its kind applied to UHI analysis in Dhaka that integrates machine learning with spatial explainability. This framework not only enhances interpretability but also offers a template for future UHI studies in other data-constrained, climate-vulnerable cities across the Global South.

2. Methodology

2.1. Study Area

Dhaka, the capital of Bangladesh, lies between 23°38'N–23°54'N latitude and 90°20'E–90°30'E longitude. The city was divided into 500 m × 500 m grid cells, a spatial resolution commonly used in urban heat studies and suitable for integrating multi-source geospatial data (Adulkongkaew et al., 2020; Lu et al., 2021; Mandal & Gnanasambandam, 2025). Characterized by a tropical monsoon climate with distinct pre-monsoon, monsoon, and winter seasons, Dhaka has experienced rapid, unplanned urban growth and significant environmental degradation. Its extreme population density and increasing exposure to heat stress make it an ideal case for exploring the spatial drivers of LST. As a representative megacity in the Global South, Dhaka offers valuable insights for developing scalable, data-driven heat mitigation strategies in similar urban contexts.

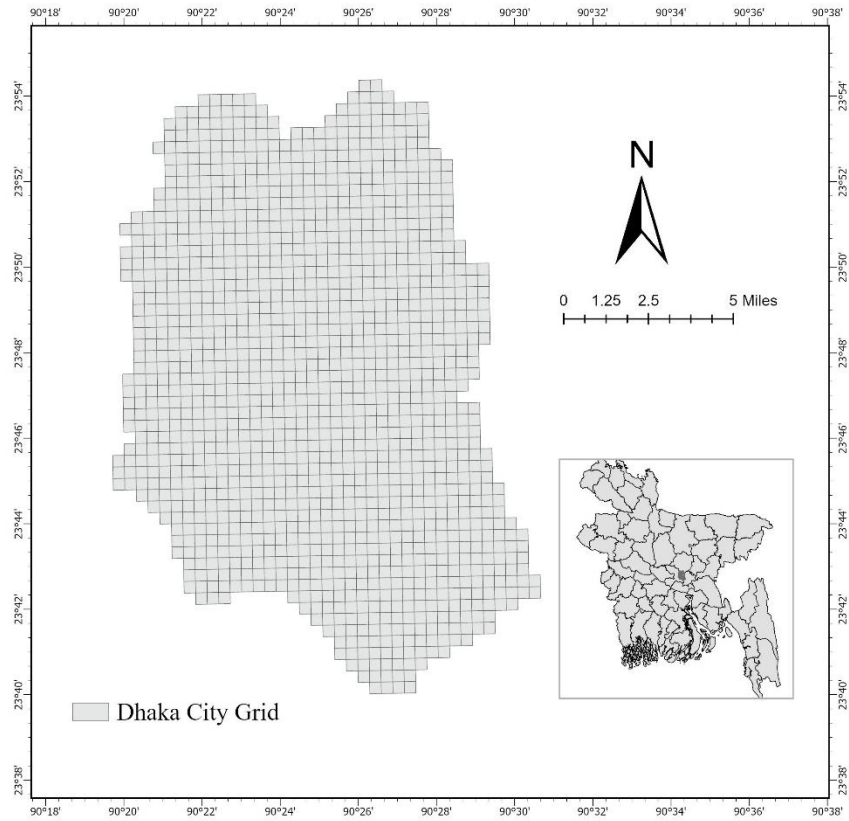


Figure 1. Geographic extent of the study area in Dhaka city, Bangladesh, subdivided into 500 m × 500 m grid cells for spatial analysis.

2.2. Data Sources and Preprocessing

This study integrates multi-source geospatial datasets, all reprojected to UTM Zone 45N (EPSG:32645) and aggregated to a standardized 500 m × 500 m grid for analysis. The primary response variable LST was derived from Landsat 8 Thermal Infrared Sensor (TIRS) Band 10. A median composite image was generated for the summer period (May to August) to avoid noise

such as cloud cover and represent seasonal thermal conditions. LST was then calculated using the mono-window algorithm, incorporating NDVI-based emissivity correction to improve accuracy in heterogeneous urban landscapes (Qin et al., 2001; Sobrino et al., 2004). The mono-window algorithm calculates LST through equation 1:

$$LST = \frac{T_B}{1 + (\lambda T_B / \rho) \ln \epsilon} \quad 1$$

where T_B represents brightness temperature, λ is the wavelength, $\rho = h \cdot c / \sigma$ (1.438×10^{-2} m K), and ϵ denotes emissivity which is calculated based on NDVI. NDVI was computed from Bands 4 (visible-red) and 5 (near infrared) of Landsat 8 OLI using the standard formula:

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad 2$$

Urban morphological and structural features were extracted from multiple geospatial data sources. Building footprints were obtained from the Google Open Buildings global dataset and aggregated within each $500 \text{ m} \times 500 \text{ m}$ grid cell to calculate the building area ratio, representing the proportion of built-up surface in each unit (Sirko et al., 2021). To account for surface composition, water bodies and bare land were identified using the Dynamic World land use/land cover (LULC) dataset (Brown et al., 2022). These classes were reclassified and processed to calculate the percentage coverage within each grid cell. Vegetation characteristics were represented by two variables: mean NDVI, derived from Landsat 8 imagery, and the proportion of tall vegetation (greater than 2 meters in height), obtained from the Meta 1-meter Canopy Height dataset (Tolan et al., 2024). The canopy height data were resampled to 10 m resolution and spatially aggregated to calculate the percentage of tall vegetation per grid. Together, these variables provide a comprehensive view of Dhaka's urban form and surface characteristics, supporting spatial modeling of LST with improved interpretability.

Table 1. Overview of datasets used in this study for LST modeling in Dhaka, Bangladesh.

Data	Variable Name	Spatial Resolution	Temporal Coverage	Source
------	---------------	--------------------	-------------------	--------

LST	Mean_LST	30 m	Apr - Aug 2020	Landsat 8 TIRS
NDVI	Mean_NDVI	30 m	Apr - Aug 2020	Landsat 8 TIRS
Building Footprints	Building_area_ratio	Vector	2020	Google Open Buildings Dataset
Canopy Height	tree_cover_gt2m_pct	10 m	2019–2020	Meta
LULC	Water_pct, bare_pct	10 m	2020	Dynamic World

2.3. Model Development and Evaluation

To identify the most suitable machine learning algorithm to model LST, this study employed an Automated Machine Learning (AutoML) framework using the FLAML library (Fast and Lightweight AutoML) developed by Microsoft (Wang et al., 2021). AutoML enables efficient model selection and hyperparameter optimization without the need for extensive manual tuning. It iteratively evaluates multiple regression algorithms and configurations to select the best-performing model within a defined computational budget.

The input feature set included key biophysical and urban morphological variables - bare land percentage, water body percentage, tree cover above 2 meters, mean NDVI, building area ratio, and spatial coordinates (X, Y). The dataset was randomly partitioned into training and testing subsets using an 80:20 split. Model training was conducted using five-fold cross-validation, a robust evaluation strategy in which the training data is split into five equal parts: in each iteration, four folds are used for training and one for validation, rotating until every fold has been used for validation once. This approach reduces overfitting and improves generalizability by ensuring that all samples are used for both training and validation.

During training, FLAML automatically explored a variety of learners including tree-based methods such as LightGBM, XGBoost, and others using an internal scoring metric (R^2 in this case) to guide optimization. Based on this iterative search, the AutoML engine selected LightGBM as the final model, owing to its efficiency and capability to handle structured tabular data with nonlinearity and feature interactions (G. Ke et al., 2017). Benchmarking studies have demonstrated that AutoML can achieve performance comparable to manually tuned models across a wide range of datasets, while substantially reducing the need for user intervention (Gijssbers et al., 2019; Wang

et al., 2021). By combining AutoML with k-fold cross-validation, this study ensures a reproducible and scalable modeling pipeline suitable for spatial prediction tasks in urban climate research.

2.4. Accuracy Assessment

To evaluate the predictive performance of the selected machine learning model, three commonly used statistical metrics were employed: Coefficient of Determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). These metrics quantify the model's ability to replicate observed LST values.

Coefficient of Determination (R^2): measures the proportion of variance in the observed LST that is explained by the model. A higher R^2 value indicates better explanatory power.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad 3$$

where y_i is the observed LST value, $\hat{y}_i$ is the predicted LST value, $\bar{y}$ is the mean of observed LST values, and n is the number of observations.

Root Mean Square Error (RMSE): quantifies the square root of the average squared differences between observed and predicted values. It penalizes larger errors more heavily and is sensitive to outliers.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad 4$$

Mean Absolute Error (MAE): represents the average of the absolute differences between observed and predicted values, offering an intuitive measure of overall prediction error.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad 5$$

Together, these metrics provide a comprehensive evaluation of model performance, capturing both the goodness of fit and the magnitude of prediction errors.

2.5. Explainable GeoAI Approach

Model interpretability was enhanced through the application of both traditional and spatially explicit explainability frameworks. First, the SHAP (SHapley Additive exPlanations) method, developed by (Lundberg & Lee, 2017b), was used to quantify the marginal contribution of each predictor to the model's LST predictions while maintaining local accuracy and consistency. SHAP assigns an important value to each feature by computing the average contribution of that feature across all possible feature combinations. SHAP values were calculated as:

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{j\}) - f(S)] \quad 6$$

where N is the set of all features, S represents a subset of features, and $f(S)$ is the model prediction for that subset.

To incorporate spatial heterogeneity, the GeoShapley method was employed as a spatially aware extension of SHAP (Li, 2024). Unlike traditional SHAP, which assumes independence among observations, GeoShapley explicitly accounts for spatial dependence and varying feature effects across geographic space. In this study, GeoShapley was applied to all grid cells, enabling a comprehensive spatial decomposition of model predictions. Each prediction was broken down into four interpretable components:

$$\hat{y}_l = \phi_0 + \phi_{GEO} + \sum_{j=1}^p \phi_j + \sum_{j=1}^p \phi(GEO, j) \quad 7$$

where ϕ_0 is the global intercept, ϕ_{GEO} represents spatial fixed effects, ϕ_j captures non-spatial feature contributions, and $\phi(GEO, j)$ accounts for spatial-feature interactions.

The resulting spatially varying coefficient (SVC) surfaces provided a nuanced understanding of how feature influence differs across geographic space. This combined SHAP and GeoShapley approach enabled both global and localized interpretation of feature importance, offering complementary insights for identifying targeted urban cooling strategies and location-specific mitigation priorities.

3. Analysis & Results

The distribution of mean LST across Dhaka exhibits a moderately right-skewed pattern, with a calculated mean of 37.75 °C and a standard deviation of 2.34 °C (Figure 2). The histogram shows that most grid cells experience LST values concentrated between 35 °C and 38 °C, indicating this as the most thermally dominant range in the city during the summer period. Although the distribution partially aligns with the overlaid normal curve, noticeable deviation is observed particularly in the right tail, where a smaller number of cells register elevated temperatures. This asymmetry suggests the presence of localized hotspots, likely driven by dense urban infrastructure, impervious surfaces, and reduced vegetation cover. The skewness indicates that the LST data does not follow a perfect normal distribution - an important consideration for statistical analyses that assume normality. The presence of both densely clustered cooler zones and sparsely distributed warmer pockets highlights the heterogeneous urban thermal environment, shaped by land cover variability and built-up density across Dhaka.

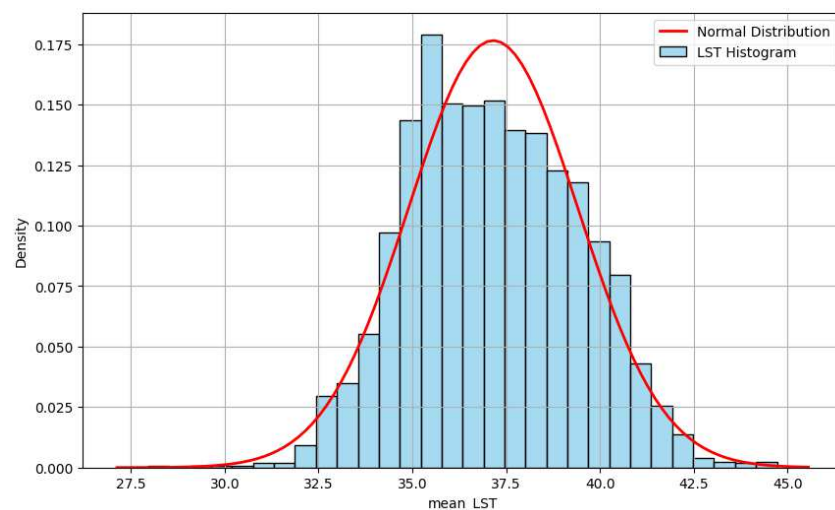


Figure 2. Histogram and fitted normal curve showing the distribution of mean LST across Dhaka.

The global Moran's I statistic for mean LST across Dhaka (Figure 3a) was found to be 0.67 with a p-value of 0.001, indicating a strong and statistically significant positive spatial autocorrelation. This means that areas with similar LST values - either high or low - tend to be spatially clustered rather than randomly distributed. The Moran's I scatterplot visually reinforces this, showing a clear linear relationship between standardized LST values and their spatial lags, where most data points fall in the first and third quadrants, corresponding to High-High and Low-Low spatial

associations. To further localize this spatial dependency, the LISA (Local Indicators of Spatial Association) cluster map (Figure 3b) identifies specific geographic patterns of LST clustering. Notably, large portions of the southern and southwestern zones of Dhaka exhibit High–High (HH) clusters, indicating concentrated urban heat hotspots. In contrast, Low–Low (LL) clusters dominate the northern and eastern peripheries, possibly associated with greener land cover or less built-up areas. Smaller High–Low (HL) and Low–High (LH) pockets are scattered throughout the city, highlighting localized thermal anomalies or edge-transition zones between contrasting thermal environments. Together, these findings emphasize the spatial heterogeneity of urban heat distribution in Dhaka and suggest that urban thermal mitigation should be geographically tailored, focusing especially on HH clusters with interventions such as urban greening, reflective surfaces, and targeted heat action planning.

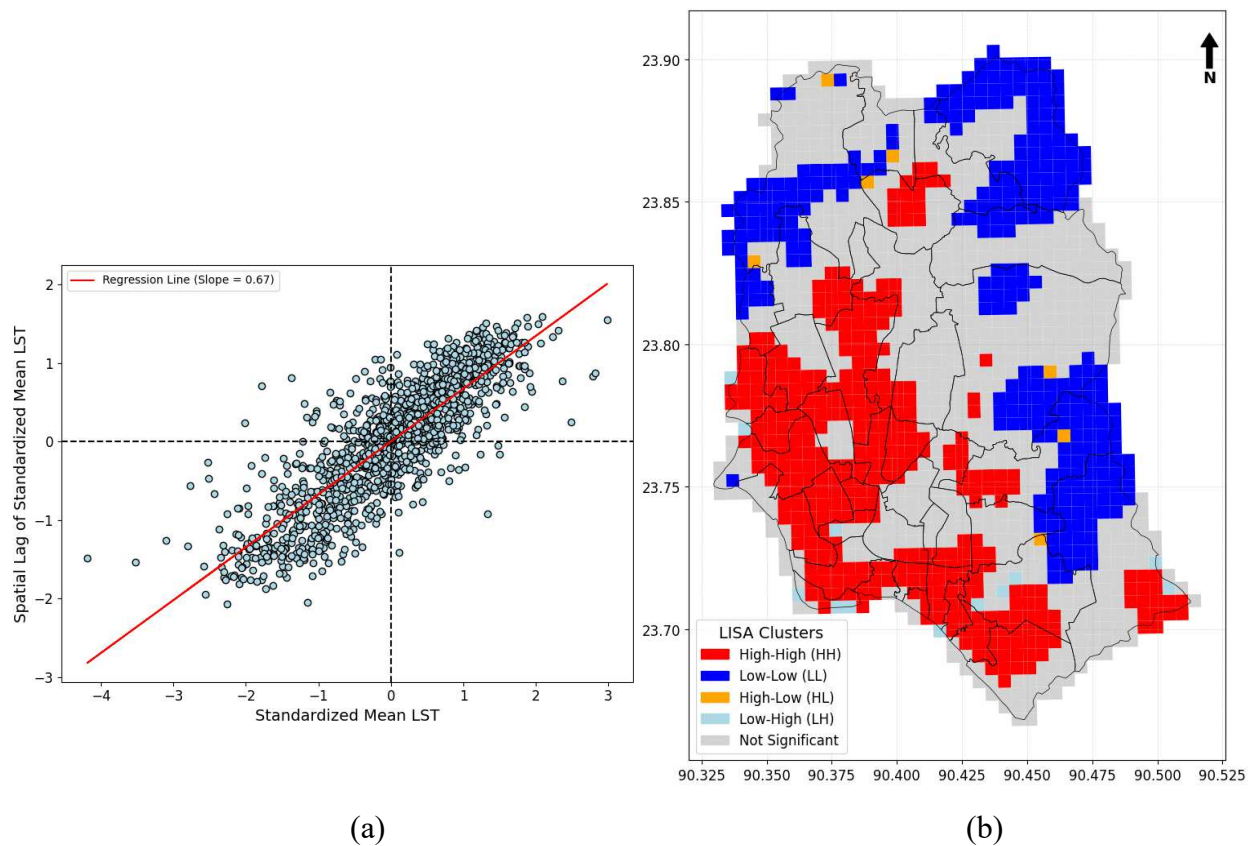


Figure 3. Spatial autocorrelation of mean LST in Dhaka: (a) Moran's I Scatterplot and (b) LISA Cluster Map.

Figure 4 illustrates the spatial distribution of six key environmental variables that influence urban thermal conditions across Dhaka. The Mean LST map reveals higher temperatures in the southern and central regions, corresponding closely with densely built-up areas. The Water Percentage layer shows sparse water coverage, with minimal influence in most of the city except for a few scattered patches in the northwest and southeast, which align with cooler LST zones. The Bare Percentage map highlights a few exposed soil areas, mostly concentrated in the north and southwest, which may contribute to local heating due to low evapotranspiration. In contrast, Mean NDVI and Tree Cover (>2m) percentages are higher in the northern and peripheral zones, indicating more vegetation, which likely contributes to the lower LST observed in those areas. Finally, the Built Area Ratio is highest in the central, southern, and eastern parts of the city, mirroring the High-High LST clusters observed in the LISA map and reinforcing the role of impervious surfaces in driving urban heat. Together, these spatial patterns demonstrate the interconnected influence of land cover, vegetation, and development intensity on the urban thermal environment in Dhaka, suggesting that green infrastructure and vegetation preservation could play a critical role in mitigating localized heat stress.

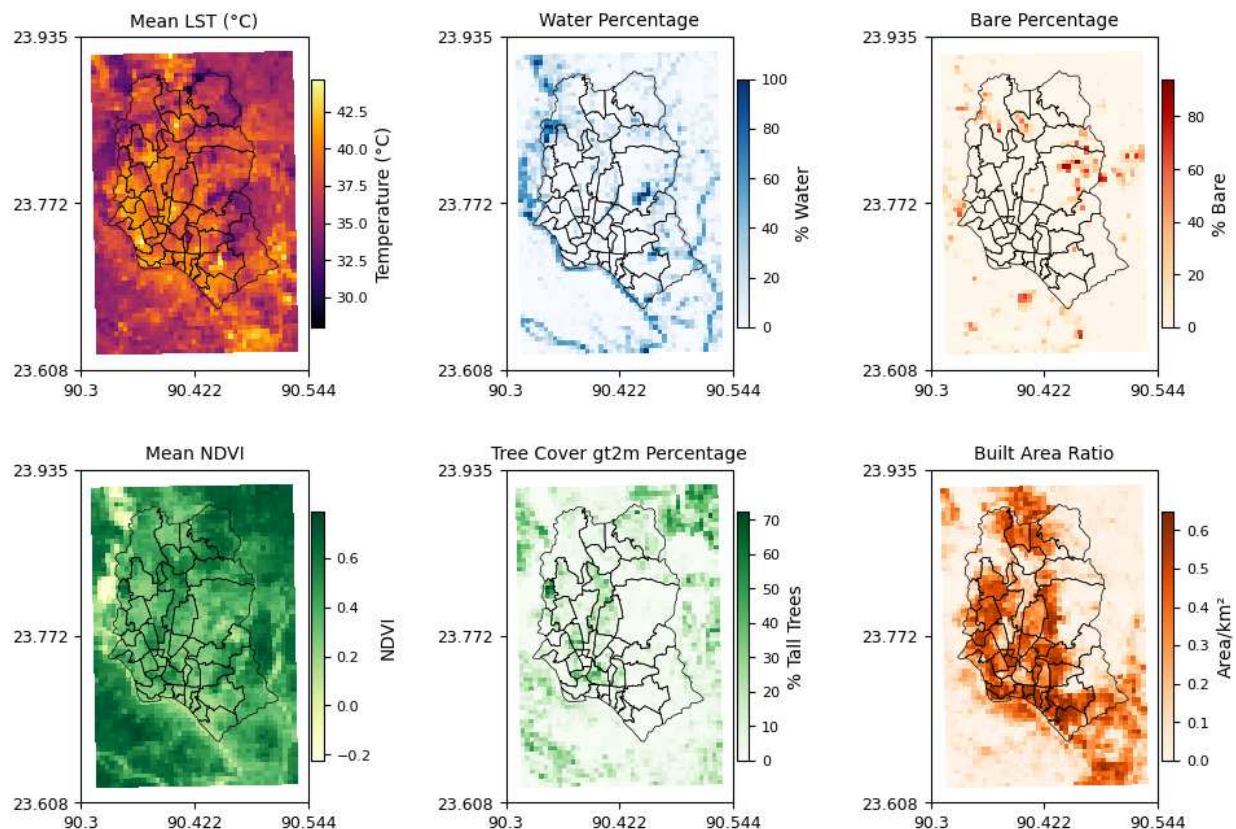


Figure 4. Spatial distribution of mean LST, water percentage, bare land percentage, mean NDVI, tree cover (>2 m), and built area ratio across Dhaka, highlighting environmental factors influencing urban thermal patterns.

The correlation matrix (Figure 5) highlights key relationships between LST and selected environmental and urban variables in Dhaka. Mean LST exhibits a strong positive correlation with built area ratio ($r = 0.62$), underscoring the role of urbanization in driving surface heating. In contrast, negative correlations with water percentage ($r = -0.45$) and mean NDVI ($r = -0.30$) suggest the cooling influence of vegetation and water bodies, while tree cover (>2m) shows a weaker negative correlation ($r = -0.04$), possibly due to sparse canopy distribution. Built area ratio also negatively correlates with NDVI ($r = -0.52$) and water percentage ($r = -0.49$), reinforcing the ecological trade-offs of urban expansion. Bare land percentage appears to have minimal influence on LST ($r = 0.02$). To ensure reliable regression modeling, Variance Inflation Factor (VIF) analysis was conducted for five core variables with all VIF values remaining below the threshold of 5. NDVI showed the highest VIF (3.47), followed by tree cover (1.89) and built area ratio (1.83), indicating no problematic multicollinearity. Together, these findings emphasize the thermal impacts of impervious surfaces and the mitigating role of green and blue infrastructure, while validating the selected predictors for modeling urban heat dynamics across Dhaka.

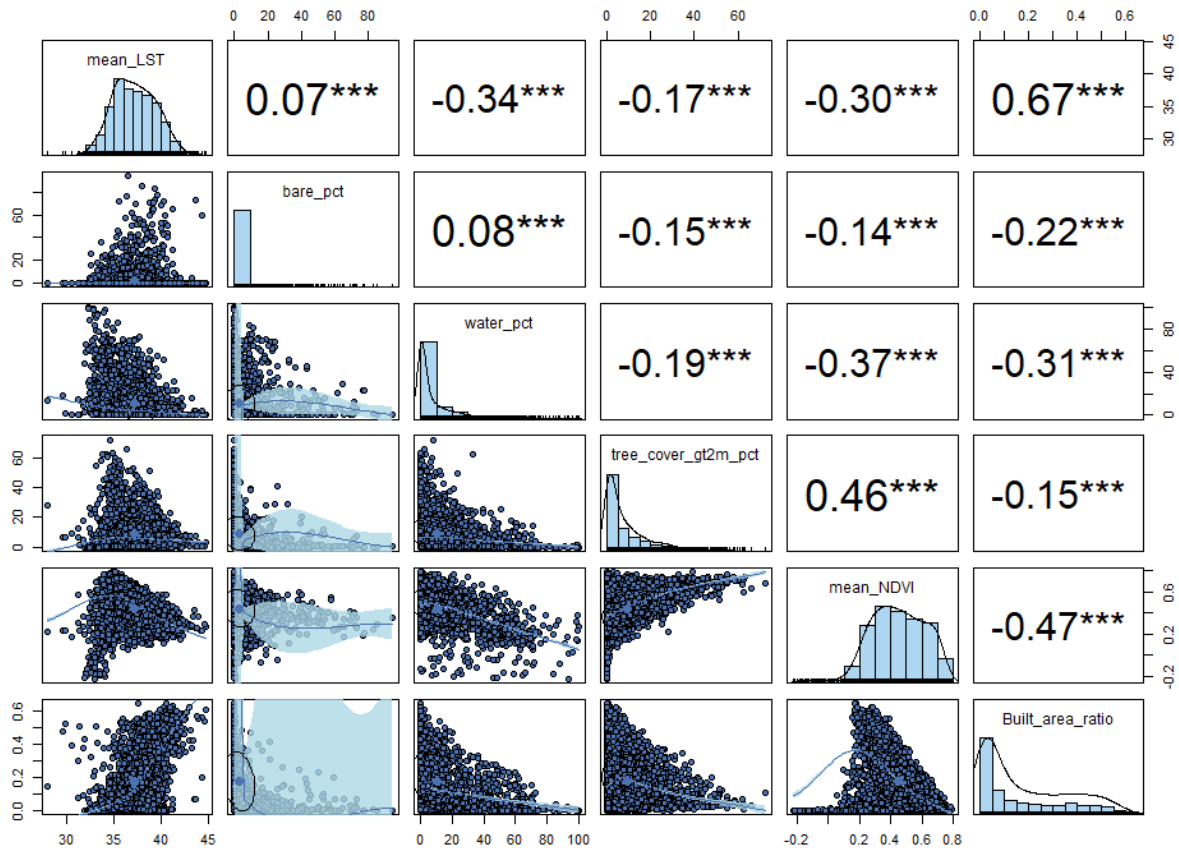


Figure 5. Correlation matrix showing the relationships between mean LST and key environmental and urban variables in Dhaka

Finally, an AutoML-based regression framework was implemented using FLAML to predict mean LST across Dhaka, leveraging key land cover and urban structure variables. FLAML automatically evaluated a range of learning algorithms under a 20-second time constraint and selected a LightGBM regressor as the optimal model based on 5-fold cross-validation performance. The final model achieved an R^2 score of 0.725, indicating that approximately 72.5% of the variance in LST was explained by these five spatial features. The RMSE was 1.18°C , and the MAE was 0.83°C , both reflecting high predictive precision. The bias was nearly negligible at -0.04°C , indicating the model did not systematically over- or under-predict. These metrics collectively validate the robustness and reliability of the LightGBM model for spatial LST prediction.

Figure 6 presents a two-part diagnostic evaluation of the model's predictive performance and spatial reliability. Figure 6a shows a scatter plot of observed versus predicted mean LST for the

test dataset, with points colored by prediction density. The alignment of most data points along the red dashed 1:1 line suggests strong model agreement, particularly in the mid-range temperatures between 34°C and 38°C, where prediction density is highest. This pattern indicates that the model consistently captures spatial variations in typical urban thermal conditions. Slight deviations at the lower and upper extremes - associated with vegetated and highly urbanized zones - reflect the increased surface heterogeneity, posing greater predictive challenges. The right panel (Figure 6b) displays Moran's I scatter plot of model residuals, where each point represents a residual and its spatial lag. The positive spatial autocorrelation (Moran's I = 0.184) suggests that the model does not leave purely random spatial errors; rather, some structured spatial patterns remain. However, the relatively modest Moran's I value indicates that the model has effectively reduced much of the spatial dependence present in the raw data. The color gradient reflects residual magnitude, with clustering of positive and negative values indicating zones of over- or underestimation. Together, these plots affirm the model's ability to not only deliver high predictive accuracy but also account for spatial dependencies, thereby supporting its use in urban heat mapping and spatially sensitive policy applications.

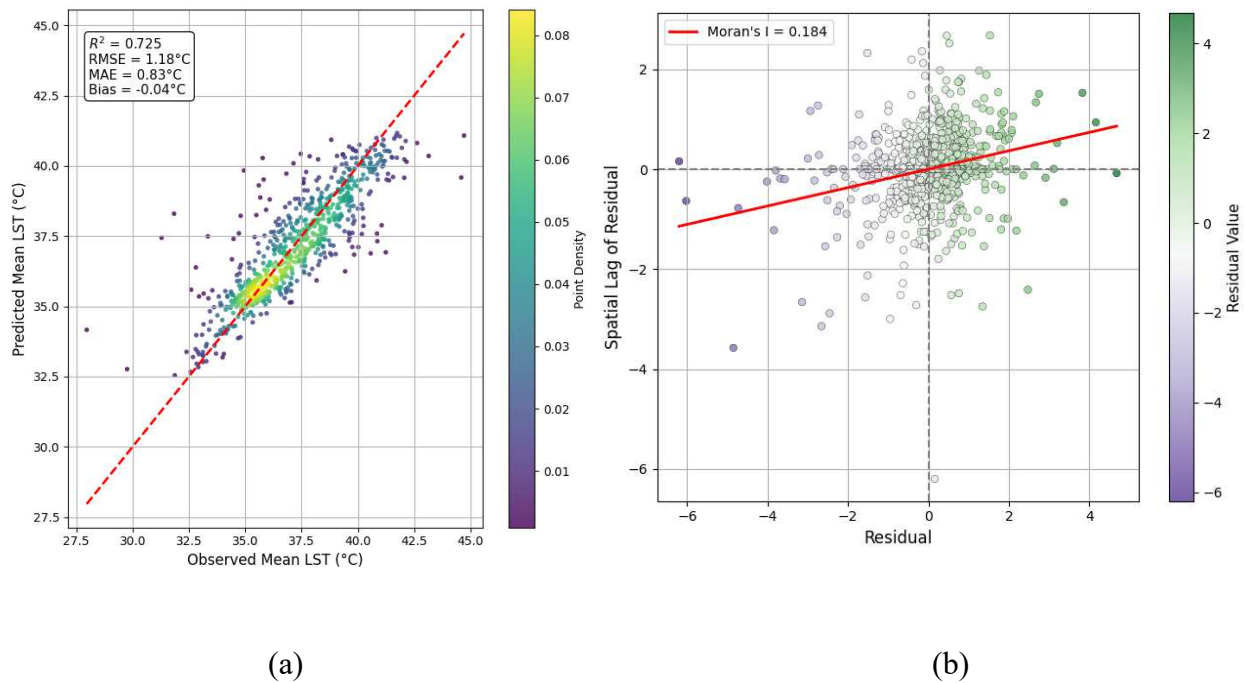


Figure 6. (a) Scatter plot of observed versus predicted mean LST values for the test dataset (b) Moran's I scatter plot of model residuals, illustrating the negligible spatial autocorrelation structure in prediction errors.

Figures 7a and 7b illustrate the global and local contributions of input features to the prediction of mean LST using the SHAP method. Figure 7a presents a SHAP beeswarm plot, where each point represents a single grid cell, with its position on the x-axis indicating the magnitude and direction of the feature's contribution to the predicted LST. Feature values are color-coded, with red representing higher values and blue representing lower values. Among all variables, Built Area Ratio emerges as the most influential feature, showing a strong and consistent positive association with LST - especially in areas with high built-up intensity, reinforcing the heat-amplifying effect of impervious surfaces. Bare land percentage also shows a generally positive impact, though with more variability in effect. Conversely, mean NDVI, water percentage, and tree cover >2m exhibit predominantly negative SHAP values for higher feature values, highlighting their role in cooling the urban environment. Figure 7b ranks the features by their mean absolute SHAP value, providing a summary of their overall importance across the dataset. Built Area Ratio ranks highest, followed by the spatial coordinates Y and X, then bare_pct, mean NDVI, and water_pct. While this reinforces the dominant influence of urban morphology on LST, the prominence of X and Y coordinates suggests that spatial location plays a role in prediction.

However, a key limitation of SHAP is its inability to interpret spatial heterogeneity. The X and Y coordinates are treated as ordinary numeric variables, lacking the ability to account for true geographic interactions or spatially varying effects. As a result, SHAP cannot explain how the influence of features like vegetation or water bodies differs across locations. This limits its usefulness for spatially explicit planning and urban heat mitigation strategies, where understanding localized effects is crucial.

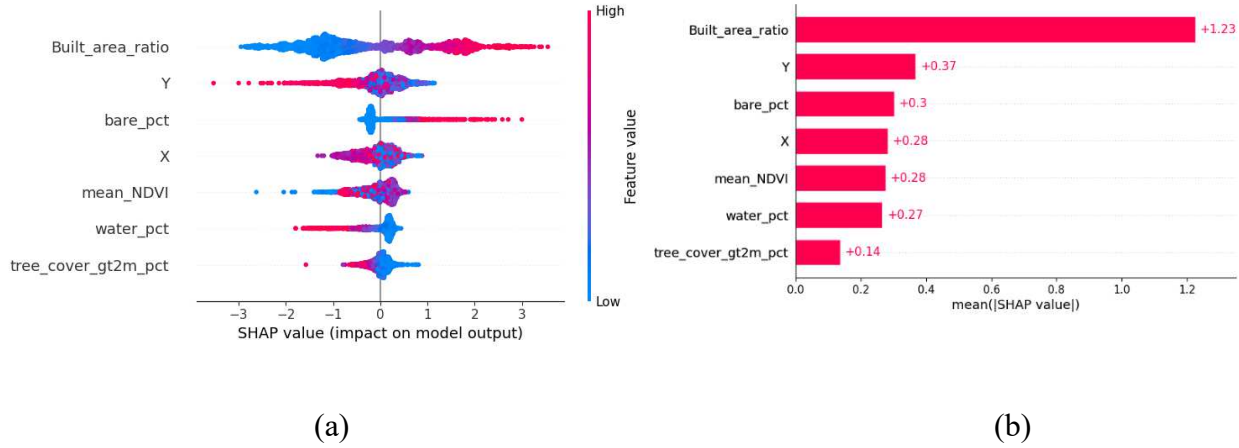


Figure 7. (a) SHAP beeswarm plot showing feature impacts on LST prediction; (b) bar plot ranking features by mean absolute SHAP values.

To overcome SHAP's limited ability to explain geographic heterogeneity in feature effects, we applied GeoShapley to produce spatially aware interpretations of LST predictions (Figure 8a and 8b). Figure 8a presents a SHAP-style summary plot using GeoShapley values, which incorporates both the global (non-spatial) and local (spatial interaction) contributions of each feature. Notably, the inclusion of spatial interaction terms (e.g., $\text{mean_NDVI} \times \text{GEO}$, $\text{bare_pct} \times \text{GEO}$, and $\text{Built_area_ratio} \times \text{GEO}$) allows for detection of spatially varying effects that standard SHAP fails to capture. For example, while mean_NDVI generally shows a negative effect on LST, the interaction term ($\text{mean_NDVI} \times \text{GEO}$) reveals that this cooling effect is not uniform across space - it intensifies or weakens depending on the location, likely due to differences in vegetation density, type, or surrounding land cover. Similarly, the spatial interaction of built_area_ratio and bare_pct shows diverse influences across regions, underscoring non-stationary relationships critical for localized planning.

Figure 8b displays the global feature importance based on the mean absolute GeoShapley values. The bar plot distinguishes between non-geo (blue) and geo-interaction (red) contributions. Built_area_ratio remains the most dominant predictor, with significant influence stemming from both its global effect and spatial interaction. Interestingly, GEO - a spatial coordinate encoding - emerges as the second-most influential variable, indicating that location itself has a meaningful impact beyond thematic features. This affirms that certain microclimatic or unmeasured locational factors (e.g., elevation, urban form, proximity to rivers) are captured through spatial encoding. The decomposition also highlights that features like mean_NDVI, bare_pct, and water_pct have considerable spatially varying effects that would be missed if only global models or SHAP were used.

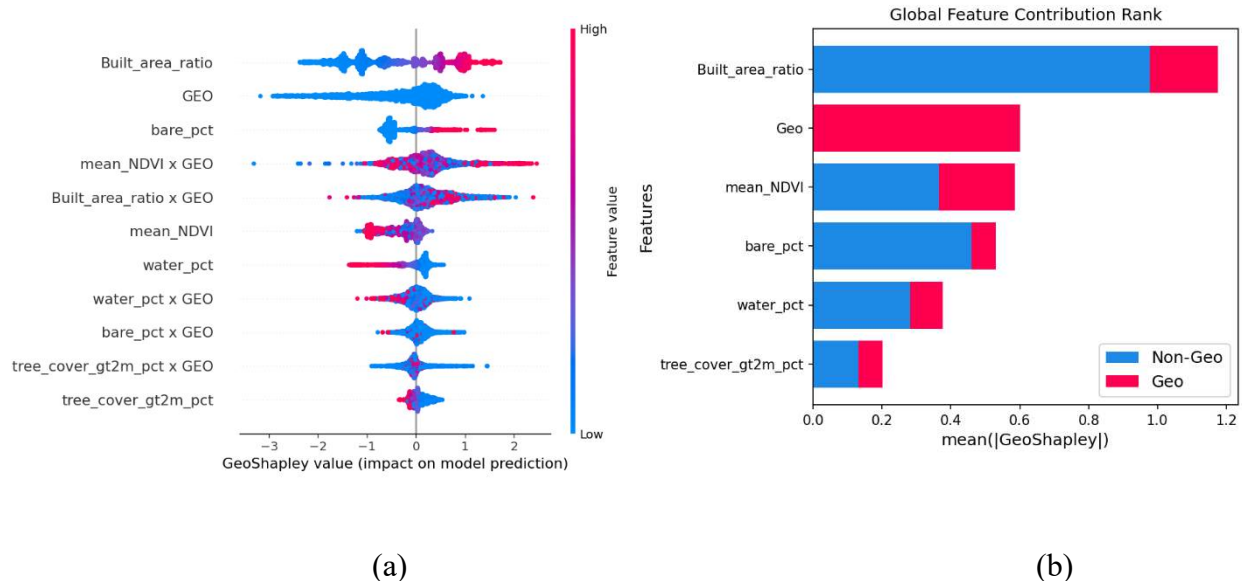


Figure 8. GeoShapley summary and feature contribution plots. (a) SHAP-style beeswarm plot showing both global and location-specific impacts of environmental predictors and their spatial interactions on LST. (b) Global ranking of features based on mean absolute GeoShapley values, decomposed into non-spatial and spatial (geo-interaction) components.

Discussion

The primary advantage of GeoShapley over SHAP lies in its capacity to explicitly quantify spatial heterogeneity and interaction effects without treating spatial coordinates merely as isolated predictors (as SHAP does with X and Y separately). By integrating game-theoretic attribution with a spatial lens, GeoShapley reveals nuanced and policy-relevant insights into how environmental drivers operate differently across space - critical for designing location-sensitive heat mitigation strategies in complex urban environments like Dhaka.

The partial dependence plots based on GeoShapley values (Figure 9) illustrate the individual and nonlinear effects of key predictors on the model's output for mean LST. Among the predictors, Built Area Ratio demonstrates the strongest and most consistent positive relationship with LST. The contribution to predicted temperature increases nearly linearly across its range, indicating that urban intensification steadily amplifies heat without clear saturation. Bare land percentage also shows a distinctly positive and nonlinear impact, with minimal contribution at very low values, followed by a sharp increase beyond approximately 10%, suggesting that exposed soil becomes a significant thermal driver in more sparsely vegetated zones. In contrast, water percentage exhibits a strong negative association with LST. As water coverage increases - particularly beyond 20% - its contribution becomes increasingly negative, confirming its critical role in cooling the urban environment. Mean NDVI, representing greenness, shows a nonlinear and inverted-U shaped effect. The GeoShapley value decreases significantly as NDVI rises from 0.2 to around 0.5, indicating substantial cooling benefits of vegetation, especially in moderately to densely green areas. However, beyond this point, the effect plateaus or slightly rebounds, suggesting diminishing returns in extremely green zones. Lastly, tree cover greater than 2 meters displays a weaker but still meaningful cooling trend. While low tree cover is associated with neutral or slightly positive thermal effects, higher values (>30%) begin to show negative contributions to LST, supporting the role of vertical green infrastructure in urban temperature moderation. Collectively, these insights highlight the complex, nonlinear, and spatially contingent influence of land cover and land use on urban heat, emphasizing the importance of targeted greening and water retention strategies in mitigating thermal stress across Dhaka.

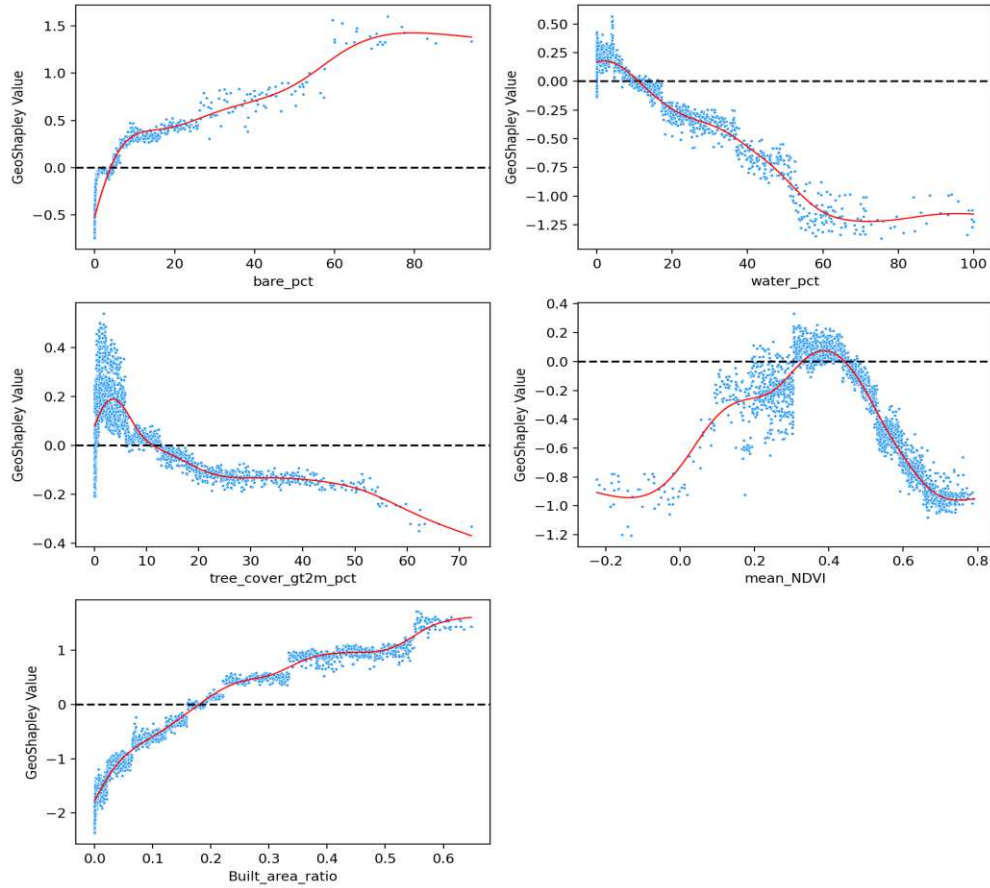


Figure 9. GeoShapley partial dependence plots showing the nonlinear effects of key environmental predictors on mean LST across Dhaka, with red curves representing smoothed trends of each variable's contribution.

To further capture the spatial complexity of these relationships, Figure 10 presents maps of both intrinsic location effects and spatially varying coefficients (SVCs) for the key predictors. The top-left map shows the intrinsic location effect, representing location-specific factors not explicitly captured by the selected features. Areas with strong negative or positive effects - depicted in red and blue - highlight underlying influences like microclimates or urban form.

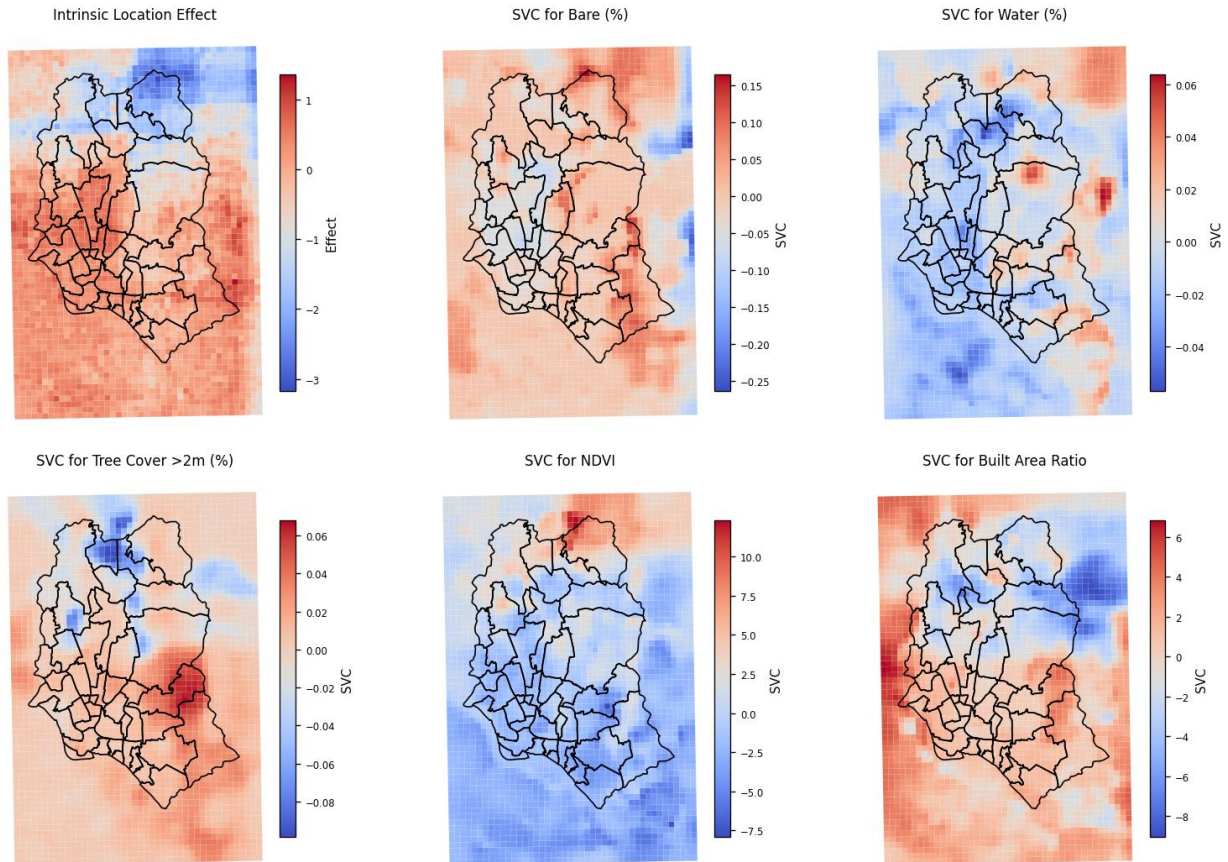


Figure 10. Spatial distribution of intrinsic location effects and SVCs for key predictors of mean LST across Dhaka, highlighting localized heterogeneity in the influence of bare land, water, tree cover, NDVI, and built area ratio.

The remaining five maps display the spatial variability of feature coefficients in the model: Bare Land Percentage, Water Percentage, Tree Cover over 2 meters, NDVI, and Built Area Ratio. These coefficients reflect how the relationship between each feature and LST varies across the city, highlighting the non-stationary nature of urban heat drivers. For instance, bare land shows a generally positive association with LST, particularly in the central and eastern regions, underscoring the heat-retaining properties of exposed soil. Water bodies exhibit a broadly negative relationship with LST, especially in the northern and southwestern zones, affirming their cooling role—though the magnitude of their effect varies spatially, likely influenced by proximity and configuration. Tree cover contributes to localized cooling, especially in the southern region, while NDVI, a proxy for overall vegetation, has a strong negative coefficient in the south and peripheral zones, suggesting the effectiveness of greenery in reducing surface temperatures. Built area ratio

consistently displays high positive coefficients in central and southern Dhaka, reflecting the intensifying heat associated with dense urban development.

These spatially varying coefficients offer critical insight into the uneven role of individual features, reinforcing the spatial heterogeneity and context dependency of LST dynamics. This study leveraged an explainable GeoAI framework to uncover these patterns, applying AutoML (FLAML) and spatially explicit model interpretation to reveal how land cover and urban morphology influence surface temperatures. While the model achieved strong predictive performance ($R^2 = 0.725$; $RMSE = 1.18^\circ\text{C}$), the deeper contribution lies in its interpretability. The intrinsic location effect maps identified residual thermal patterns in the northern and central parts of Dhaka, areas where unobserved influences such as urban geometry, anthropogenic heat, or microclimates may intensify or mitigate exposure. Together, these findings affirm that urban heat drivers are not spatially uniform and that planning responses must be localized, data-driven, and sensitive to intra-urban environmental variation.

Conclusion

The integration of explainable GeoAI techniques has proven effective in disentangling the spatial complexity of urban heat drivers in Dhaka. This study not only produced accurate LST predictions but also unveiled how the effects of urban features vary across space. These insights are not accessible through traditional global models. The results underscore that green infrastructure, tree canopy preservation, and strategic placement of water features can be particularly effective in mitigating urban heat, but their benefits are location specific. Moreover, the significant intrinsic location effects observed point to the influence of unmeasured or hidden urban factors, such as wind corridors, surface albedo, and building configurations, which future research should aim to incorporate. From a planning perspective, these findings provide a valuable decision-support tool for designing targeted urban cooling strategies in heat-prone zones. By prioritizing greening efforts and reducing impervious surface concentration in high-SVC areas, urban policymakers can more equitably and effectively respond to climate-induced thermal stress. This work thus contributes not only to academic understanding of UHI dynamics in a South Asian megacity context but also to the advancement of spatially tailored, data-driven planning for climate resilience.

References

- Abrar, R., Sarkar, S. K., Nishtha, K. T., Talukdar, S., Shahfahad, Rahman, A., Islam, A. R. M. T., & Mosavi, A. (2022). Assessing the Spatial Mapping of Heat Vulnerability under Urban Heat Island (UHI) Effect in the Dhaka Metropolitan Area. *Sustainability*, 14(9), Article 9. <https://doi.org/10.3390/su14094945>
- Adulkongkaew, T., Satapanajaru, T., Charoenhirunyingyos, S., & Singhirunnusorn, W. (2020). Effect of land cover composition and building configuration on land surface temperature in an urban-sprawl city, case study in Bangkok Metropolitan Area, Thailand. *Heliyon*, 6(8), e04485. <https://doi.org/10.1016/j.heliyon.2020.e04485>
- Ahmed, B., Kamruzzaman, M., Zhu, X., Rahman, M. S., & Choi, K. (2013). Simulating Land Cover Changes and Their Impacts on Land Surface Temperature in Dhaka, Bangladesh. *Remote Sensing*, 5(11), Article 11. <https://doi.org/10.3390/rs5115969>
- Bappa, S. A., Malaker, T., Mia, M. R., & Islam, M. D. (2022). Spatio-temporal variation of land use and land cover changes and their impact on land surface temperature: A case of Kutupalong Refugee Camp, Bangladesh. *Heliyon*, 8(9). <https://doi.org/10.1016/j.heliyon.2022.e10449>
- Brown, C. F., Brumby, S. P., Guzder-Williams, B., Birch, T., Hyde, S. B., Mazzariello, J., Czerwinski, W., Pasquarella, V. J., Haertel, R., Ilyushchenko, S., Schwehr, K., Weisse, M., Stolle, F., Hanson, C., Guinan, O., Moore, R., & Tait, A. M. (2022). Dynamic World, Near real-time global 10 m land use land cover mapping. *Scientific Data*, 9(1), 251. <https://doi.org/10.1038/s41597-022-01307-4>
- Chen, R., Wang, C., Que, X., Liao, F. H., Ma, X., Wang, Z., Li, Z., Wen, K., Lai, Y., & Xu, X. (2024). Exploring Urban Heat Distribution and Thermal Comfort Exposure Using Spatiotemporal Weighted Regression (STWR). *Buildings*, 14(6), Article 6. <https://doi.org/10.3390/buildings14061883>
- Chen, Y., Ye, Y., Liu, X., Yin, C., & Jones, C. A. (2025). Examining the nonlinear and spatial heterogeneity of housing prices in urban Beijing: An application of GeoShapley. *Habitat International*, 162, 103439. <https://doi.org/10.1016/j.habitatint.2025.103439>

- Cleland, S. E., Steinhardt, W., Neas, L. M., Jason West, J., & Rappold, A. G. (2023). Urban heat island impacts on heat-related cardiovascular morbidity: A time series analysis of older adults in US metropolitan areas. *Environment International*, 178, 108005.
<https://doi.org/10.1016/j.envint.2023.108005>
- Feng, F., Ren, Y., Xu, C., Jia, B., Wu, S., & Laforteza, R. (2024). Exploring the non-linear impacts of urban features on land surface temperature using explainable artificial intelligence. *Urban Climate*, 56, 102045. <https://doi.org/10.1016/j.uclim.2024.102045>
- Foroutan, E., Hu, T., & Li, Z. (2025). Revealing key factors of heat-related illnesses using geospatial explainable AI model: A case study in Texas, USA. *Sustainable Cities and Society*, 122, 106243.
<https://doi.org/10.1016/j.scs.2025.106243>
- Gijsbers, P., LeDell, E., Thomas, J., Poirier, S., Bischl, B., & Vanschoren, J. (2019). *An Open Source AutoML Benchmark* (No. arXiv:1907.00909). arXiv. <https://doi.org/10.48550/arXiv.1907.00909>
- Hu, Y., Wu, C., Meadows, M. E., & Feng, M. (2023). Pixel level spatial variability modeling using SHAP reveals the relative importance of factors influencing LST. *Environmental Monitoring and Assessment*, 195(3), 407. <https://doi.org/10.1007/s10661-023-10950-2>
- Imran, H. M., Hossain, A., Islam, A. K. M. S., Rahman, A., Bhuiyan, M. A. E., Paul, S., & Alam, A. (2021). Impact of Land Cover Changes on Land Surface Temperature and Human Thermal Comfort in Dhaka City of Bangladesh. *Earth Systems and Environment*, 5(3), 667–693.
<https://doi.org/10.1007/s41748-021-00243-4>
- Islam, M. D., Islam, K. S., Ahasan, R., Mia, M. R., & Haque, M. E. (2021). A data-driven machine learning-based approach for urban land cover change modeling: A case of Khulna City Corporation area. *Remote Sensing Applications: Society and Environment*, 24, 100634.
<https://doi.org/10.1016/j.rsase.2021.100634>

- Islam, M. D., Li, B., Islam, K. S., Ahasan, R., Mia, Md. R., & Haque, M. E. (2022). Airbnb rental price modeling based on Latent Dirichlet Allocation and MESF-XGBoost composite model. *Machine Learning with Applications*, 7, 100208. <https://doi.org/10.1016/j.mlwa.2021.100208>
- Islam, M. D., Li, B., Lee, C., & Wang, X. (2022). Incorporating spatial information in machine learning: The Moran eigenvector spatial filter approach. *Transactions in GIS*, 26(2), 902–922. <https://doi.org/10.1111/tgis.12894>
- Kafy, A.-A., Naim, Md. N. H., Subramanyam, G., Faisal, A.-A., Ahmed, N. U., Rakib, A. A., Kona, M. A., & Sattar, G. S. (2021). Cellular Automata approach in dynamic modelling of land cover changes using RapidEye images in Dhaka, Bangladesh. *Environmental Challenges*, 4, 100084. <https://doi.org/10.1016/j.envc.2021.100084>
- Ke, E., Zhao, J., & Zhao, Y. (2025). Investigating the influence of nonlinear spatial heterogeneity in urban flooding factors using geographic explainable artificial intelligence. *Journal of Hydrology*, 648, 132398. <https://doi.org/10.1016/j.jhydrol.2024.132398>
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems*, 30. https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html
- Kim, M., Kim, D., & Kim, G. (2022). Examining the Relationship between Land Use/Land Cover (LULC) and Land Surface Temperature (LST) Using Explainable Artificial Intelligence (XAI) Models: A Case Study of Seoul, South Korea. *International Journal of Environmental Research and Public Health*, 19(23), Article 23. <https://doi.org/10.3390/ijerph192315926>

- Kulsum, U., & Moniruzzaman, M. (2022). Exploring the relationship of climate change and land-use dynamics with satellite-derived surface indices and temperature in greater Dhaka, Bangladesh. *Journal of Earth System Science*, 131(2), 117. <https://doi.org/10.1007/s12040-022-01841-0>
- Li, Z. (2024). GeoShapley: A Game Theory Approach to Measuring Spatial Effects in Machine Learning Models. *Annals of the American Association of Geographers*, 114(7), 1365–1385. <https://doi.org/10.1080/24694452.2024.2350982>
- Lu, X.-Y., Chen, X., Zhao, X.-L., Lv, D.-J., & Zhang, Y. (2021). Assessing the impact of land surface temperature on urban net primary productivity increment based on geographically weighted regression model. *Scientific Reports*, 11(1), 22282. <https://doi.org/10.1038/s41598-021-01757-7>
- Lundberg, S., & Lee, S.-I. (2017a). *A Unified Approach to Interpreting Model Predictions* (No. arXiv:1705.07874). arXiv. <https://doi.org/10.48550/arXiv.1705.07874>
- Lundberg, S., & Lee, S.-I. (2017b). *A Unified Approach to Interpreting Model Predictions* (No. arXiv:1705.07874). arXiv. <https://doi.org/10.48550/arXiv.1705.07874>
- Mandal, G., & Gnanasambandam, S. (2025). Temporal impact of urban indicators of land surface cover on land surface temperature: Case study of an Indian metropolitan city. *Environment and Planning B: Urban Analytics and City Science*, 52(2), 286–302. <https://doi.org/10.1177/23998083241256497>
- Morshed, N., Yorke, C., & Zhang, Q. (2017). Urban Expansion Pattern and Land Use Dynamics in Dhaka, 1989–2014. *The Professional Geographer*, 69(3), 396–411. <https://doi.org/10.1080/00330124.2016.1268058>
- Pande, C. B., Egbueri, J. C., Costache, R., Sidek, L. M., Wang, Q., Alshehri, F., Din, N. M., Gautam, V. K., & Chandra Pal, S. (2024). Predictive modeling of land surface temperature (LST) based on Landsat-8 satellite data and machine learning models for sustainable development. *Journal of Cleaner Production*, 444, 141035. <https://doi.org/10.1016/j.jclepro.2024.141035>

- Peng, Z., Ji, H., Yuan, R., Wang, Y., Easa, S. M., Wang, C., Cui, H., & Zhao, X. (2025). Modeling and spatial analysis of heavy-duty truck CO₂ using travel activities. *Journal of Transport Geography*, 124, 104158. <https://doi.org/10.1016/j.jtrangeo.2025.104158>
- Qin, Z., Karnieli, A., & Berliner, P. (2001). A mono-window algorithm for retrieving land surface temperature from Landsat TM data and its application to the Israel-Egypt border region. *International Journal of Remote Sensing*, 22(18), 3719–3746. <https://doi.org/10.1080/01431160010006971>
- Rahman, M. M., Avtar, R., Yunus, A. P., Dou, J., Misra, P., Takeuchi, W., Sahu, N., Kumar, P., Johnson, B. A., Dasgupta, R., Kharrazi, A., Chakraborty, S., & Agustiono Kurniawan, T. (2020). Monitoring Effect of Spatial Growth on Land Surface Temperature in Dhaka. *Remote Sensing*, 12(7), Article 7. <https://doi.org/10.3390/rs12071191>
- Rihan, M., Talukdar, S., Naikoo, M. W., Islam, M. R., Shahfahad, & Rahman, A. (2025). Optimizing Land Use/Land Cover Mapping Accuracy: A Comprehensive Analysis of Random Forest with Explainable Artificial Intelligence Based Feature Importance Analysis. *Journal of the Indian Society of Remote Sensing*. <https://doi.org/10.1007/s12524-025-02229-5>
- Saiful Islam, K., Islam, M. D., Chakraborty, T., & Alam, Md. S. (2019). *URBAN HEAT ISLAND EFFECT ANALYSIS USING INTEGRATED GEOSPATIAL TECHNIQUES: A CASE STUDY ON KHULNA CITY, BANGLADESH*.
- Sheng, T., Zhang, Z., Qian, Z., Ma, P., Xie, W., Zeng, Y., Zhang, K., Sun, Z., Yu, J., & Chen, M. (2025). Examining urban agglomeration heat island with explainable AI: An enhanced consideration of anthropogenic heat emissions. *Urban Climate*, 59, 102251. <https://doi.org/10.1016/j.uclim.2024.102251>

- Si, M., Li, Z.-L., Tang, B.-H., Liu, X., & Nerry, F. (2024). Spatial heterogeneity of driving factors-induced impacts for global long-term surface urban heat island. *International Journal of Remote Sensing*, 45(19–20), 7139–7159. <https://doi.org/10.1080/01431161.2023.2203343>
- Sirko, W., Kashubin, S., Ritter, M., Annkah, A., Bouchareb, Y. S. E., Dauphin, Y., Keysers, D., Neumann, M., Cisse, M., & Quinn, J. (2021). *Continental-Scale Building Detection from High Resolution Satellite Imagery* (No. arXiv:2107.12283). arXiv. <https://doi.org/10.48550/arXiv.2107.12283>
- Sobrino, J. A., Jiménez-Muñoz, J. C., & Paolini, L. (2004). Land surface temperature retrieval from LANDSAT TM 5. *Remote Sensing of Environment*, 90(4), 434–440. <https://doi.org/10.1016/j.rse.2004.02.003>
- Tabassum, A., Hong, S.-H., Park, K., & Baik, J.-J. (2025). Simulating urban heat islands and local winds in the Dhaka metropolitan area, Bangladesh. *Urban Climate*, 59, 102284. <https://doi.org/10.1016/j.uclim.2024.102284>
- Tobler, W. R. (1970). A Computer Movie Simulating Urban Growth in the Detroit Region. *Economic Geography*, 46, 234–240. <https://doi.org/10.2307/143141>
- Tolan, J., Yang, H.-I., Nosarzewski, B., Couairon, G., Vo, H. V., Brandt, J., Spore, J., Majumdar, S., Haziza, D., Vamaraju, J., Moutakanni, T., Bojanowski, P., Johns, T., White, B., Tiecke, T., & Couprie, C. (2024). Very high resolution canopy height maps from RGB imagery using self-supervised vision transformer and convolutional decoder trained on aerial lidar. *Remote Sensing of Environment*, 300, 113888. <https://doi.org/10.1016/j.rse.2023.113888>
- Uddin, A. S. M. S., Khan, N., Islam, A. R. Md. T., Kamruzzaman, M., & Shahid, S. (2022). Changes in urbanization and urban heat island effect in Dhaka city. *Theoretical and Applied Climatology*, 147(3), 891–907. <https://doi.org/10.1007/s00704-021-03872-x>
- Voogt, J. A., & Oke, T. R. (2003). Thermal remote sensing of urban climates. *Remote Sensing of Environment*, 86(3), 370–384. [https://doi.org/10.1016/S0034-4257\(03\)00079-8](https://doi.org/10.1016/S0034-4257(03)00079-8)

Wang, C., Wu, Q., Weimer, M., & Zhu, E. (2021). *FLAML: A Fast and Lightweight AutoML Library* (No. arXiv:1911.04706). arXiv. <https://doi.org/10.48550/arXiv.1911.04706>

Wei, J., Li, Y., Jia, L., Liu, B., & Jiang, Y. (2025). The Impact of Spatiotemporal Effect and Relevant Factors on the Urban Thermal Environment Through the XGBoost-SHAP Model. *Land*, 14(2), Article 2. <https://doi.org/10.3390/land14020394>

Wu, R., Yu, G., & Cao, Y. (2025). The impact of industrial structural transformation in the Yangtze River economic belt on the trade-offs and synergies between urbanization and carbon balance. *Ecological Indicators*, 171, 113165. <https://doi.org/10.1016/j.ecolind.2025.113165>